Rohit Raj

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EDUCATION

Juniata College, Huntingdon, PA — May 2025
Bachelor of Science, Physics — *Graduated with Distinction*
Studied abroad: University of Leeds, United Kingdom, 2023–2024

Teaching: TA and evening group tutor for Intro Physics I & II, Modern Physics, and E&M (Fall 2022, Spring 2023, Fall 2024, Spring 2025), Juniata College

AWARDS & SCHOLARSHIPS

- **Chambliss Astronomy Achievement Student Award**, American Astronomical Society — January '26
  Awarded for exemplary research quality and presentation skill at the 247th AAS.
- **Outstanding Undergraduate Research Award**, American Institute of Physics — May '25
  One of two students nationally recognized for exceptional undergraduate research.
- **Herbert Levy Memorial Scholarship**, American Institute of Physics — May '24
  One of two national awardees for academic excellence and pursuit of high scholarship.
- **Rufus Reber Physics Prize**, Juniata College — May '24
  Awarded for outstanding academic achievement and citizenship.
- **SPS–Google Scholarship**, Google — May '23
  One of twenty students selected nationally for academic potential, resilience, and demonstrated ability to overcome significant obstacles.
- **FAMOUS Travel Grant**, American Astronomical Society — December '23
  Competitive travel award supporting early–career researchers presenting at AAS.
- **SPS Research Reporter Award**, American Physical Society — November '23
  Selected to report scientific sessions for SPS national publications.
- **PhysCon Underrepresented Student Travel Award**, American Institute of Physics — September '22
  Fully funded to present research at Physics Congress 2022.
- **Frontline Worker Scholarship**, Outlier.org — March '21
  Awarded for community service efforts during the COVID–19 pandemic.
- **Next Genius Scholarship**, India — January '21
  One of 187 scholars selected from 4,500 applicants for merit–based tuition support.

RESEARCH EXPERIENCE

Center for Astrophysics | Harvard & Smithsonian — September '25 – Present

- **SPARK Postbac Fellow:** Modeled the surface mass ejection driving Betelgeuse's 'Great Dimming' using the Rybicki-Hummer non-LTE radiative transfer code; reproduced the observed Mg II chromospheric line asymmetries and quantified the ejection velocity and mass-loss rate, consistent with HST/STIS UV time-domain spectra. First-author paper in preparation. *Advisors: Andrea Dupree, Paul Cristofari.*
- Building a neural spectral emulator to forward-model the effect of magnetic spot distributions on cool star spectra, with direct applications to next-generation high-resolution spectrographs.

University of California, Santa Cruz (UCSC) — June 2024 – August 2024

- **SIP Research Lead:** Led spectroscopic and photometric analysis of Wolf-Rayet (WR) and unusual emission-line stars in M31, M32, and M33 using Keck/DEIMOS and HST PHATTER/PHAST photometry; mentored three high school interns. *Advisors: Raja GuhaThakurta, Bob Williams.*
- Developed a 2D Gaussian crowding analysis pipeline using HST photometry; recovered sub-arcsecond ($< 0.1''$) coordinates for $> 90\%$ of the M33 WR sample and demonstrated that blending in Hectospec-scale fiber surveys systematically biases WR subtype classifications and mass-loss rate estimates derived from equivalent widths. *ApJ* paper in review; **Keck OSIRIS LGSAO follow-up awarded (2026B, Program U221)**.

University of California, Santa Cruz (UCSC) — June 2023 – August 2023

- **SIP Research Lead:** Identified systematic velocity suppression in DESI fiber spectroscopy relative to DEIMOS slit spectroscopy in M31's crowded disk, revealing precision limits of $1.5''$-fiber spectrographs for resolving individual stellar kinematics. Presented at 243rd AAS. *Advisor: Raja GuhaThakurta.*
- Developed an HST-based crowding contamination metric ranking blending severity across survey targets; this analysis directly motivated the 2D Gaussian crowding pipeline applied in the subsequent M33 WR study.

Imperial College, London, UK (Remote) August 2024 – December 2024

- **Theoretical Physics Research Student:** Developed FDTD simulations in Python to analyze EM wave propagation in media with time-varying permittivity, including optical cloaking in quasi-crystals; reproduced results from seminal literature. *Advisor: T. V. Raziman.*

Emory University, Laney Graduate School (Remote) May '23 – August '23

- **SOAR Scholar:** Conducted a theoretical physics reading project focused on classical and quantum wave equations, analyzing applications of Noether's theorem to the Classical and Schrödinger Wave Equations.

Case Western Reserve University, Cleveland May 2022 – August 2022

- **Research Intern:** Implemented matched filtering and Fourier analysis to reproduce LIGO's GW150914 binary black hole detection; developed educational materials on gravitational wave data analysis. *Mentor: Harsh Mathur.*

PUBLICATIONS

- **Raj, R.**, Cristofari, P.I., MacLeod, M., Huang, C., Dupree, A.K. "Modeling the Chromospheric Signature of a Mass Ejection in Betelgeuse." *The Astrophysical Journal*, in prep. (co-author review; anticipated submission August 2026).
- **Raj, R.**, Guhathakurta, P., Williams, R.E., Williams, B.F., Dalcanton, J.J., Durbin, M.J. "Wolf-Rayet Stars in M33: The Impact of Stellar Crowding on Spectral Classification and Mass-Loss Diagnostics." *The Astrophysical Journal*, submitted (July 2026).

OBSERVING PROPOSALS

- **Co-Investigator** (proposal author), Keck Observatory OSIRIS LGSAO, Program U221 Awarded 2026B
  "Wolf-Rayet Star Environments in M33: Resolving Spectral Contamination with K-band AO IFU Spectroscopy"
  PI: P. GuhaThakurta (UCSC) · 1 night (2 half-nights), Keck I / OSIRIS LGSAO, November 2026

PRESENTATION

- *"A Source-Grounded AI Workflow for Accessible Astronomy Outreach,"* Poster, Language AI in the Space Sciences Workshop at Space Telescope Science Institute (STScI), Baltimore, MD, **March, 2026**.
- *"Modeling the Chromospheric Dynamics of Betelgeuse during the Great Dimming,"* Poster, 247^{th} Meeting of the American Astronomical Society **(AAS)**, Phoenix, AZ, **January, 2026**.
- *"Wolf-Rayet Stars in M33 and a Few Other Unusual Emission-Line Stars in M31, M32, and M33,"* Poster, Sigma Pi Sigma Physics Congress, Denver CO, **October, 2025**.
- *"Wolf-Rayet Stars in M33 and a Few Other Unusual Emission-Line Stars in M31, M32, and M33,"* Poster 353.18, 245^{th} Meeting of the American Astronomical Society **(AAS)**, National Harbour, MD, **January, 2025**.
- *"Comparing SPLASH and DESI Surveys for Insights into M31's Evolution,"* Poster 405.04, 243^{rd} Meeting of the American Astronomical Society **(AAS)**, New Orleans, LA, **January, 2024**.
- *"Simulating the Kinematics of Globular Clusters,"* **Talk**, Liberal Arts Symposium, Juniata College, PA, **April 2023**.
- *"Searching for Black Holes in LIGO data,"* Poster, Sigma Pi Sigma Physics Congress, Washington DC, **October, 2022** & **Talk**, 13th Annual Landmark Conference, Moravian College, PA, **July, 2022**.
- *"Non–linear Dynamics — Chaotic Double Pendulum,"* **Talk**, Liberal Arts Symposium, Juniata College, PA, **April, 2022**.

TRAINING PROGRAM

NASA Proposal Writing and Evaluation Experience — Co-Author September '24 – December '24

- Co-authored a comprehensive technical proposal for NASA Marshall Space Flight Center, serving as Student Chief Scientist for a multidisciplinary team. Participated in weekly sessions led by scientists, mastering proposal writing.
- Served as a primary reviewer on a NASA review panel, gaining hands-on experience in evaluation methodologies and mission design.

Canadian Astroparticle Physics Summer School — Participant May '23

- Selected for a competitive, funded intensive training program at Queen's University. Participated in hands-on dark matter detection workshops and received training on the Nobel Prize-winning SNO experiment at SNOLAB.

Perimeter Institute, Canada — Participant in Gravity and Black Hole Workshop March '21

OUTREACH & COMMUNITY ENGAGEMENT

Scientific Outreach

Cambridge Explores the Universe, Cambridge, MA October '25

- Managed the Micro Observatory station and served as an Ask-an-Astronomer representative.

Cambridge Science Festival, Cambridge, MA September '25

- Assisted with experiment setup and operations; supported hands-on physics demonstrations and public engagement.

SPS and Sigma Pi Sigma Council, American Institute of Physics July '24 – June '25

- Led the student engagement efforts in the Northeast US, including organizing annual regional physics department meetings. Proposed communication strategies to the AIP Board to improve undergraduate involvement.

The Business Show, London, UK October '24

- Featured as a researcher on a live radio segment (London/Spotify), communicating astrophysics research and advocating for STEM access and EDI initiatives.

Shadow the Scientist, UC Santa Cruz, CA November '22

- Coordinated with Creating Equity in STEAM (UCSC) and Dr. Nazma to deliver an "Astronomy Night at W.M. Keck Observatory" outreach event for students in a terrorism-affected region of Kashmir, India.

Society of Physics Students, Juniata College, PA August '22 – May '25

- Social Media Officer: Expanded department visibility on LinkedIn, advertised outreach events, and produced science communication content connecting students, alumni, and the community.

U.S.–India Cultural Exchange

SPAN Magazine, U.S. Embassy New Delhi, India November '23

- Provided insights to Indian students on navigating U.S. research internships and graduate opportunities.

U.S.–India Dosti Youth Ambassador 2023

- Promoted cross-cultural educational and scientific exchange initiatives between India and the United States.

SPAN Magazine, U.S. Embassy New Delhi, India November '22

- Featured in "Looking Beyond Ivy Leagues," advocating a broad, inclusive view of global educational success.

Leadership in Social Justice

COVID–19 Relief Coordination, India 2020 – 21

- Led nationwide remote coordination of relief logistics by mobilizing student volunteers, donors, and government officials. Arranged long-distance transport (special trains and buses) for stranded migrant workers and coordinated food distribution along transit routes.
- Facilitated delivery of essential supplies—oxygen cylinders, medicines, and ration kits—supporting several thousand individuals.

Non-Profit Teaching Center, Begusarai, Bihar March '20 – Present

- Founded and manage a free learning center serving 50 underprivileged students; supervise three teachers and integrate AI-generated regional-language STEM materials to support accessible instruction.

Zomato Feeding, Bhubaneswar, India May '19 – March '21

- Established partnerships with local restaurants to direct surplus food donations to underserved communities; recognized with a certificate for serving over 2000 meals.

SKILLS

Programming: Python (Astropy, NumPy, SciPy, Matplotlib, pandas, Dask); R; C++ *[intermediate]*
Machine Learning & Emulation: JAX/Flax, PyTorch, Scikit-learn; neural spectral emulators, Bayesian inference
Astrophysical Tools: Radiative transfer modeling (Rybicki-Hummer non-LTE); IRAF, DS9; spectral fitting, photometric reduction, image processing
Data & Surveys: FITS format, large survey pipelines (DESI, Keck/DEIMOS, HST PHATTER/PHAST), crowding analysis
High-Performance Computing: SLURM job scheduling, parallel computing, GPU workflows
Languages: English *[fluent]*, Hindi *[fluent]*